



PPM

Infotech

Real-time industry experience....



FULL STACK

WEB DEVELOPMENT

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COURSE OBJECTIVE

Our objective is to provide freshers with practical, industry-focused training in modern web development technologies, where PPM Infotech serves as the platform for learning, innovation, and real-time project experience. Through expert mentorship, hands-on exposure to frontend, backend, database, and AI-based development tools, we aim to transform students into skilled, confident, and job-ready professionals with valuable industry experience and performance-based certification.

KEY HIGHLIGHTS OF THE TRAINING

- 🎯 **Complete fulstack development**
- 🎯 **Expert Trainer Support**
- 🎯 **Real Work Experience**
- 🎯 **personal doubt solving**

- 🎯 **Upgrade Your Career in 6 Months.**
- 🎯 **Live Project Training**
- 🎯 **Work Experience Certificate**
- 🎯 **Real-time industry experience**

Over view and tools coverd in the course

Webdevelopment

Frontent



HTML



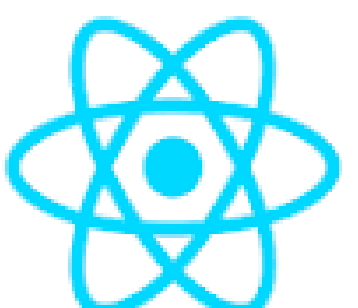
CSS



JavaScript



Bootstrap



React

Backend

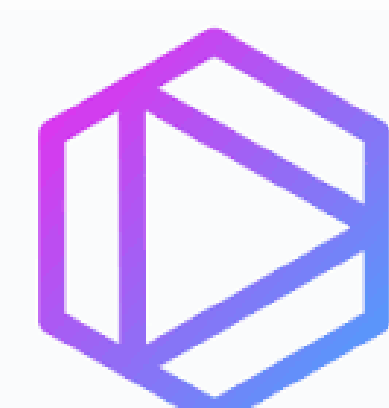
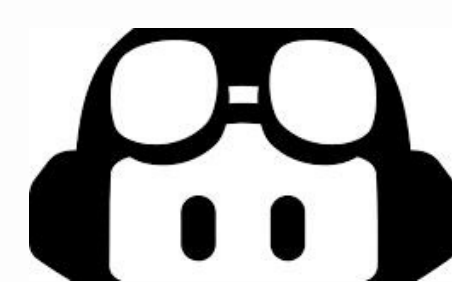
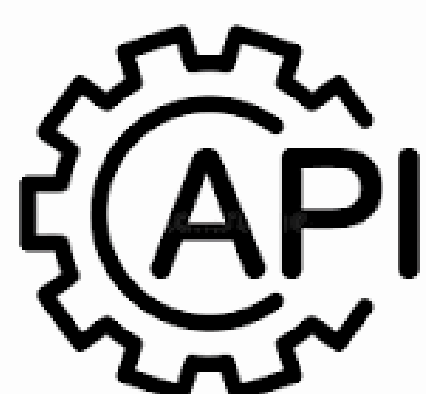


Express JS

Database



Core Foundations & Career Enhancement Modules



Frontend Development

Module 1

Getting into Coding with VS Code & Version Control Systems

Programming & Development Basics	Programming intro, SDLC overview, workflow basics
VS Code Setup & Interface	Installation, extensions, themes, explorer, terminal
Project & File Management	Folder structure, file creation, workspace setup
Command Line & Git Fundamentals	Terminal commands, navigation, Git basics, workflow
Git Commands & GitHub	Init, add, commit, push, pull, clone, branching, merge
First Project Execution	Run code, environment setup, basic testing

Module 2

Designing Web Pages with HTML, css & Bootstrap

HTML Basics	Inline elements, Block elements, Element attributes
HTML Forms	HTML forms and form elements, Types of inputs, Buttons
HTML Media & Links	Anchor tags, Images, Video
HTML Meta & Utility Tags	Meta tags, Miscellaneous tags (code, table, span, etc.)
CSS Fundamentals	Cascading, CSS selectors
CSS Selectors	Element selectors, Pseudo selectors, Attribute selectors
CSS Colors & Backgrounds	CSS colors, Color picker, Gradients, Background properties
CSS Box Model	Box model, Padding, Margin, Border
CSS Layout & Alignment	CSS alignment, Flexbox, Grid, Positioning
Advanced CSS	CSS transformations, iFrames, Animations, Webkit & keyframes

Bootstrap(Responsive UI Development)

Bootstrap Basics	Bootstrap introduction, paragraphs, alignments
Contextual Styling	Contextual classes, Labels
Buttons & Forms	Buttons, Links, Submit buttons, Forms
Grid System	Bootstrap grid system and responsive layouts

Module 3

Making Websites Dynamic with JavaScript

JavaScript Programming

JavaScript Fundamentals (ES6)	Variables, data types, operators, control flow
Functions & OOP	JavaScript functions, Arrow functions, Classes, OOP concepts
DOM Manipulation	DOM access, DOM manipulation
Events	DOM events and event handling
Higher Order Functions	forEach, map, filter, reduce
JavaScript Best Practices	Tips and tricks for clean and efficient JS code

Module 4

Building Powerful User Interfaces with React.js

React.js Devlopment

React Basics	Creating React app, Folder structure
Components	Understanding components, Functional components, Class components
JSX & Events	JSX syntax, Event handling
Props & State	Working with props, State management
Forms & Inputs	Handling user inputs, Adding forms

Lifecycle & Hooks	Component lifecycle, React Hooks (useState, useEffect)
HTTP Communication	Making HTTP requests using Axios
Routing	Routing in React
Context & State Management	React Context API
Advanced State Management	Redux, Redux middleware (Thunk, Logger)
Third-party Integrations	Using third-party APIs

Front End Development Project

Frontend Capstone Project	End-to-end frontend application using HTML, CSS, JavaScript, React
API Integration Project	Integrating third-party APIs
State Management Implementation	Redux-based state management
Deployment Readiness	Project structure, optimization, best practices

“Complete Frontend → Build Live Projects → Move to Backend Development.”

Backend Development

Module 1

Node.Js

Node.js Introduction	What is Node.js, V8 engine, Node vs browser JS, Blocking vs non-blocking, Event loop overview
Node Runtime & Globals	global object, process object, _ dirname / environment variables, CLI arguments
Module System	require(), module.exports, import/export, CommonJS vs ES Modules, npm & package.json
Core Modules	fs module (read/write), path module, os module, events module, synchronous vs async APIs

File System Deep Dive	reading files async, writing files, append, streams basics, buffers intro
HTTP Server	http module, request lifecycle, routing manually, headers, status codes
Async Programming	callbacks, promises, async/await, error handling, async flow control
Events & Streams	EventEmitter, custom events, streams types, piping, backpressure concept
Node Architecture	single thread model, worker threads intro, libuv overview, concurrency vs parallelism, scalability concepts
Mini Project	file-based API, routing, async FS, JSON handling, testing endpoints

Module 2

Express.js Introduction

Express Intro	What is Express, Setup server, Routing basics, req/res, JSON responses
Routing System	GET/POST
Routing System	PUT/DELETE
Data management	route params, query params, REST conventions, route organization

Middleware

Middleware fundamentals	middleware flow, next(), built-in middleware, express.json, static middleware
Custom Middleware	logger middleware, request timing, reusable middleware, middleware chaining, global vs route middleware

MVC Structure

Project architecture	MVC pattern, controllers
Project architecture	Routes folder, services layer, config structure

Validation & Error Handling

Validation libraries	validation concept, schema validation, sanitization, error responses, reusable validators
Error Handling	try/catch in async, next(error), global error handler, custom errors, response format

Api's

REST principles	resources, HTTP verbs, status codes, statelessness, naming conventions
CRUD API	create resource, read resource, update resource, delete resource, testing endpoints
Advanced REST	pagination, filtering, sorting, search, field selection

Authentication & Authorization

Authentication Intro	authentication vs authorization, sessions vs tokens, login flow, protected routes, roles
Password Security	hashing concept, bcrypt usage, salting, password storage, verify passwords
JWT Authentication	JWT structure, signing token, verifying token, expiry, payload
Auth Middleware	token extraction, verify middleware, user injection, protected endpoints, role middleware

Authorization	RBAC concept, role middleware, permission checks, admin routes, access control
Security Libraries	CORS, Helmet, rate limiting, XSS protection, request limits
File Upload	multipart data, upload middleware, storage options, file validation, serving files
API Documentation	API docs concept, endpoint description, request/response schema, auth docs, testing tools
Testing Express API	endpoint testing, auth testing, error testing, integration tests, test structure

Module 3

MONGODB Introduction

Database Intro	SQL vs NoSQL, Mongo architecture, collections, documents, BSON
Mongo Setup	Mongo server, shell, connection, databases, collections

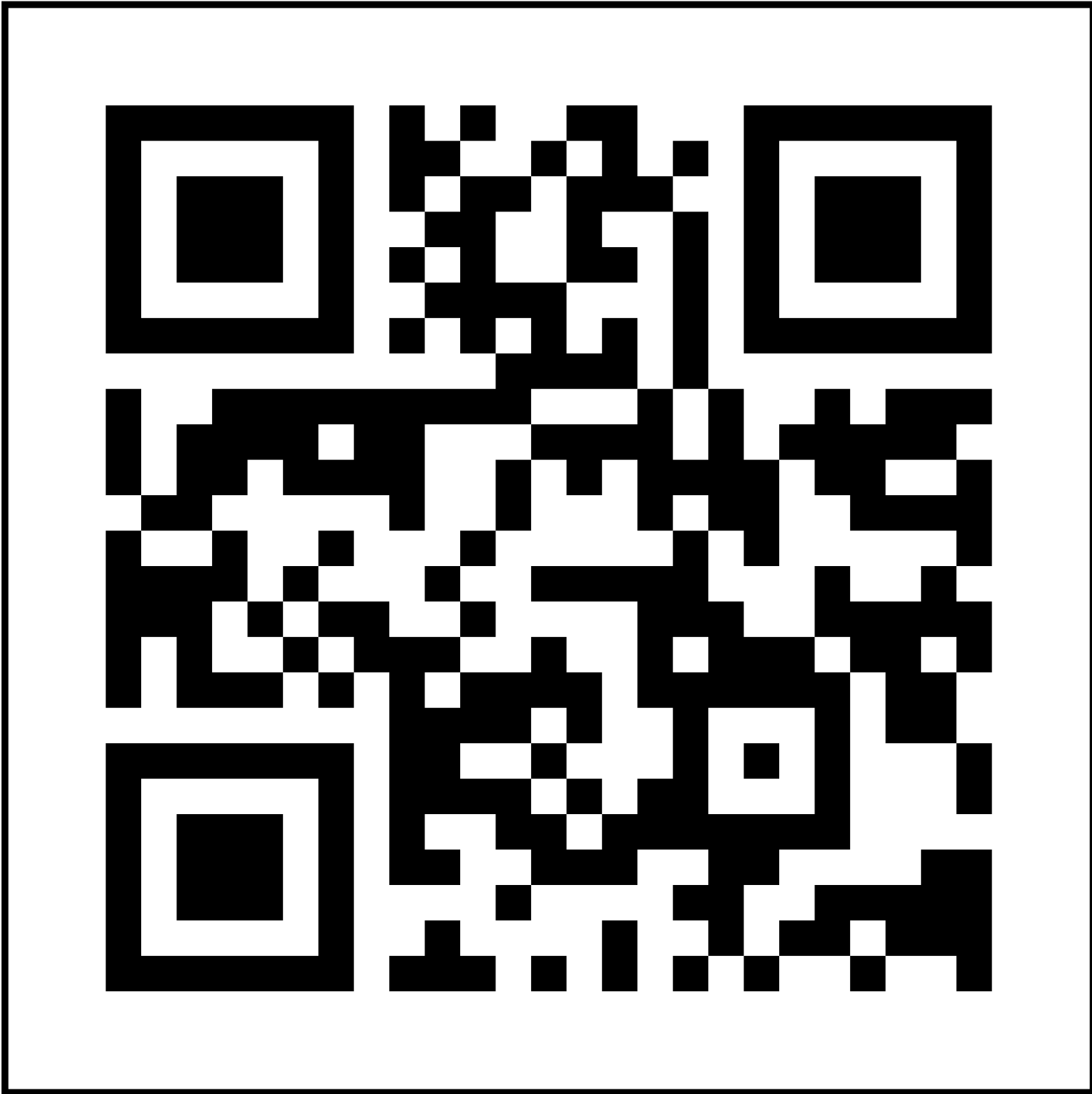
CRUD

Create & Read	insertOne insertMany, document structure, find, filters, projections
Update & Delete	updateOne/updateMany, \$set \$inc, array updates, deleteOne/deleteMany, safe updates
Query Operators	comparison operators (\$gt, \$lt, \$in), logical operators (\$and, \$or), element operators (\$exists, \$type), regex search, nested queries
Arrays & Embedded Docs	arrays in MongoDB, array queries, \$push / \$pull, embedded documents, dot notation

Advanced

Relationships & Schema Design	embedding vs referencing, ObjectId references, one-to-many, many-to-many, schema design patterns
Indexing & Performance	index concept, single & compound index, unique index, query performance, explain()
MongoDB with Node & Project	MongoDB driver, connection string, CRUD from Node, error handling, mini data model project

Course Fee & Offer Details

Course Fee	Scan to visit our website
AI-integrated Full Stack Development Program ✖ 35,000/- ✔ 30,000/-	

Best of Luck!



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